Achouri Youcef Islem

Applied AI and Machine Learning Internship

Sidi Bel Abbès, Algeria · yi.achouri@esi-sba.dz
github.com/youcef-ach · linkedin.com/in/youcef-achouri

Final-year engineering student focused on computer vision and NLP. Available January 2027.

EDUCATION

ESI-SBA 2021 - Present

École Nationale Supérieure d'Informatique de Sidi Bel Abbès
State Engineering Degree in Computer Science - fifth year, degree in progress.
Specialty: Artificial Intelligence and Data Structures (AIDS).

SELECTED PROJECTS

Spatial Computer Vision | Python, PyTorch, Babylon.js

- Built a pipeline that segments windows and doors in panoramic imagery using Mask2Former and SAM, then projects those masks onto 3D wall geometry.
- Implemented ray-plane lifting and depth checks for occlusion, with a WebGPU viewer for inspecting reconstructed scenes and lighting.

Arabic Structured Information Extraction | Python, Qwen 2.5, Ollama 2025

- Authored the NLP module for the team's PlanMilt DZ academic GIS project, converting Arabic descriptions into structured scene data.
- Defined the extraction schema and prompts, with JSON parsing, validation and synthetic Arabic examples for local evaluation.

Neural Network from Scratch | Python, NumPy 2026

- Implemented a two-layer MNIST classifier with vectorized forward propagation, analytical backpropagation and mini-batch gradient descent, without automatic differentiation.

English to French Translation | PyTorch 2026

- Implemented Transformer attention, causal masking and positional encodings, with a custom byte-pair tokenizer and training pipeline.

TECHNICAL SKILLS

Machine learning: Python, PyTorch, NumPy, neural networks, model evaluation.
Computer vision: Mask2Former, SAM, panoramic imagery, 2D-to-3D projection.
NLP: Transformer attention, byte-pair tokenization, Qwen 2.5, Ollama, JSON Schema.
Engineering: FastAPI, Docker, PostgreSQL, Git, pytest, Linux.

LEADERSHIP AND COMMUNICATION

Vice President | INGENIUMS Club, ESI-SBA 2023 - 2024

- Co-organized student technology events and programming contests, including Ingeneer, Ingehack and Ingecode.
- Spoke at ESI-SBA Welcome Day and demonstrated technical projects at campus exhibitions.

AWARDS

2nd place, DevFest Hackathon 2023 - GDG Sidi Bel Abbès, Team Infinity.
3rd place, NCS Hackathon 2024 - Numidia Computer Society / NiT.

Languages: English and French (strong proficiency).